Objective: Containerize a full-stack application — a Node.js/Express frontend and a Flask backend — using Docker and Docker Compose, then push the images to Docker Hub and the code to GitHub.

1. Frontend (Node.js with Express)

1. Create a frontend using Express and Node.js.
2. Include a form similar to the one from the Flask Assignment 2.
3. Configure the form to send a request to the Flask backend.

2. Backend (Flask)

4. Use the Flask backend to handle the form submission and process the data.

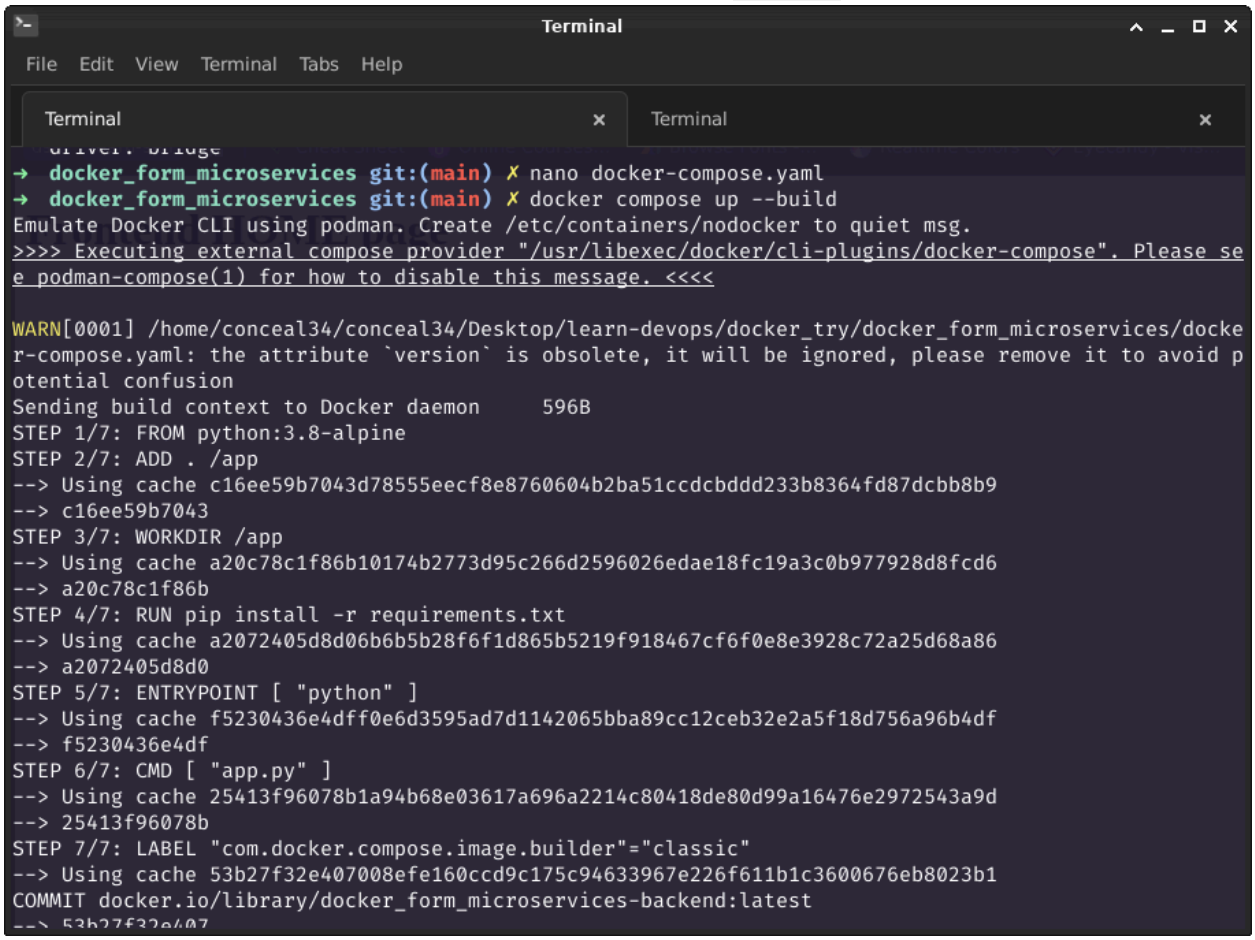
3. Folder Structure

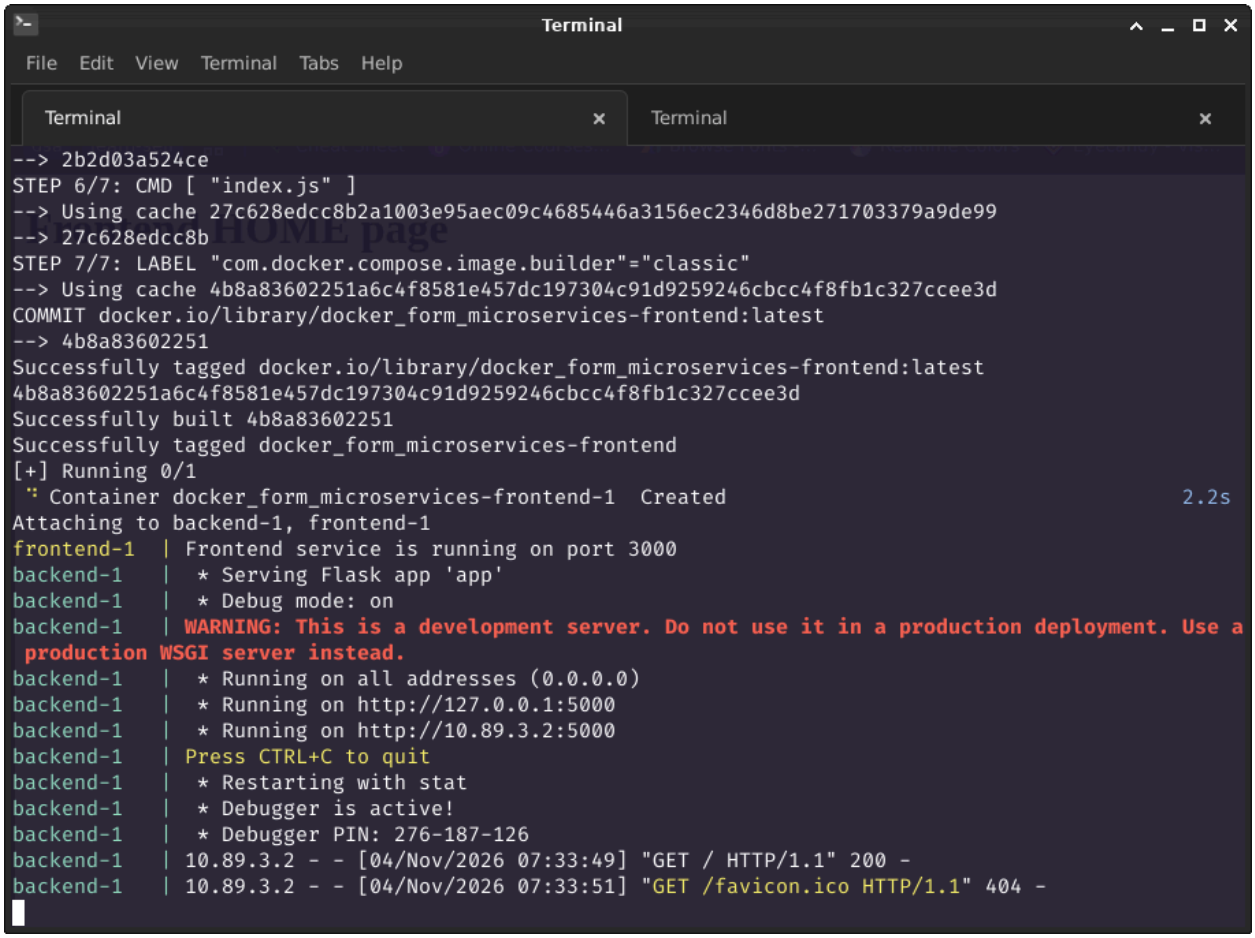
5. Organize the project with separate folders for the frontend and backend.

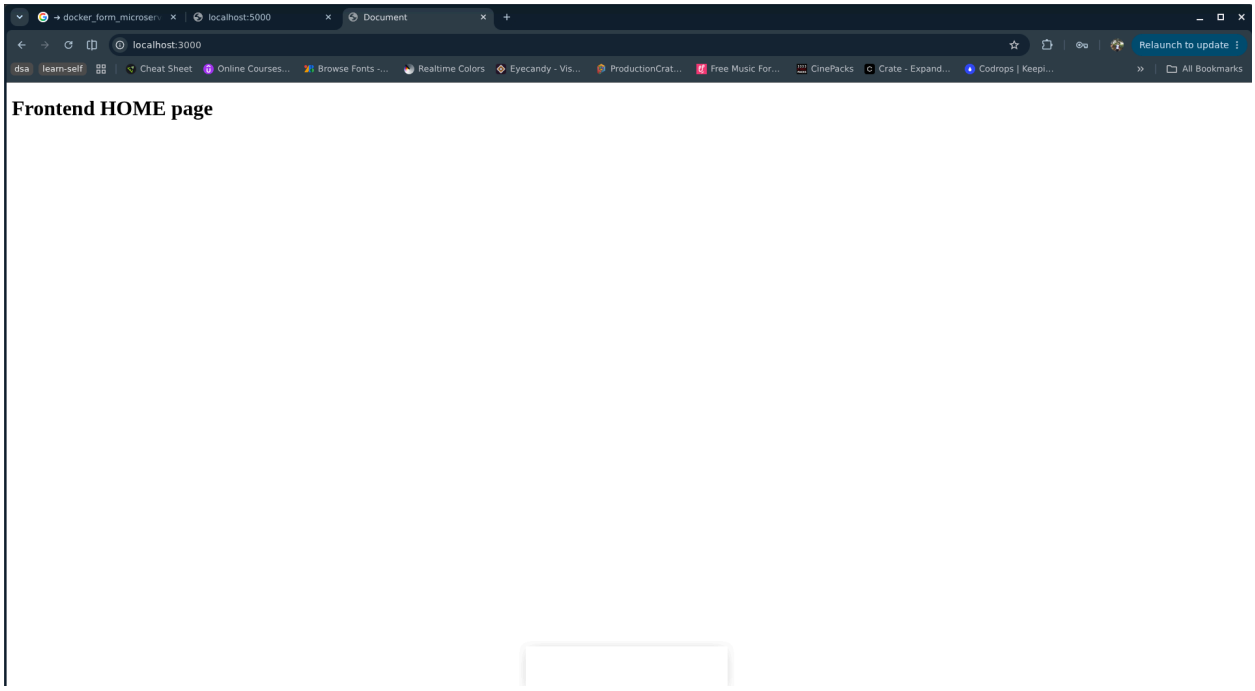
4. Docker Configuration

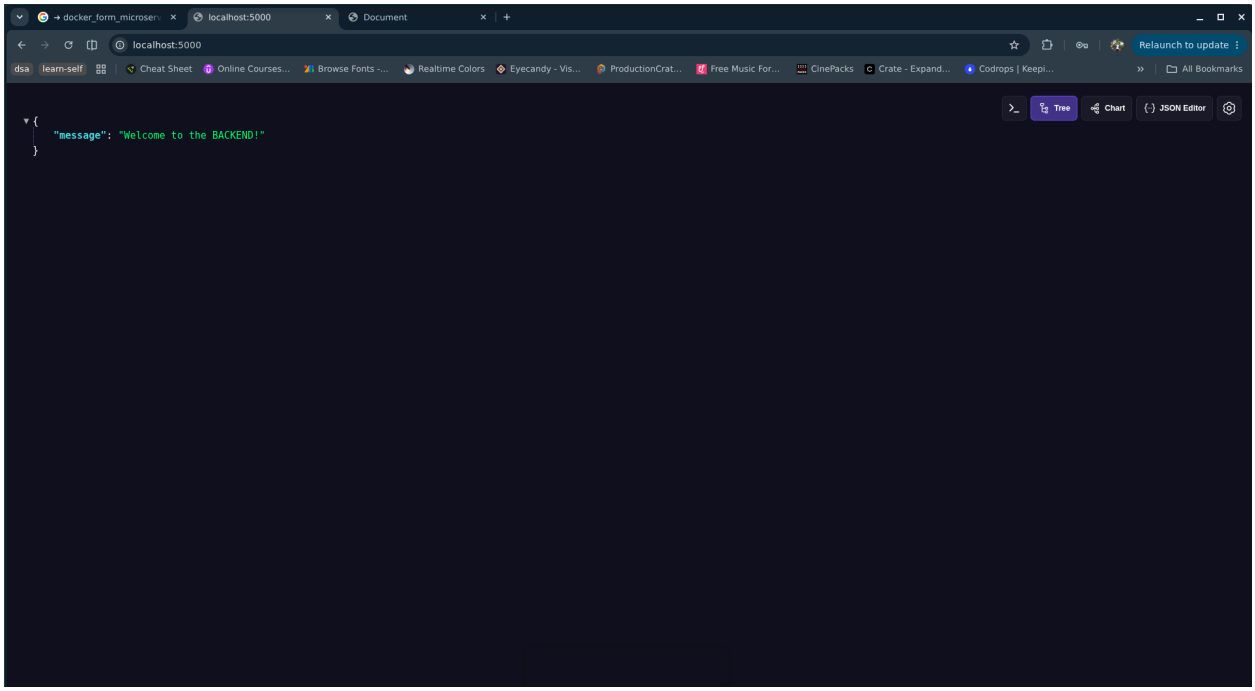
6. Create a `Dockerfile` for both the frontend and backend.
7. Write a `.yaml` file (Docker Compose) to connect both services in the same network.

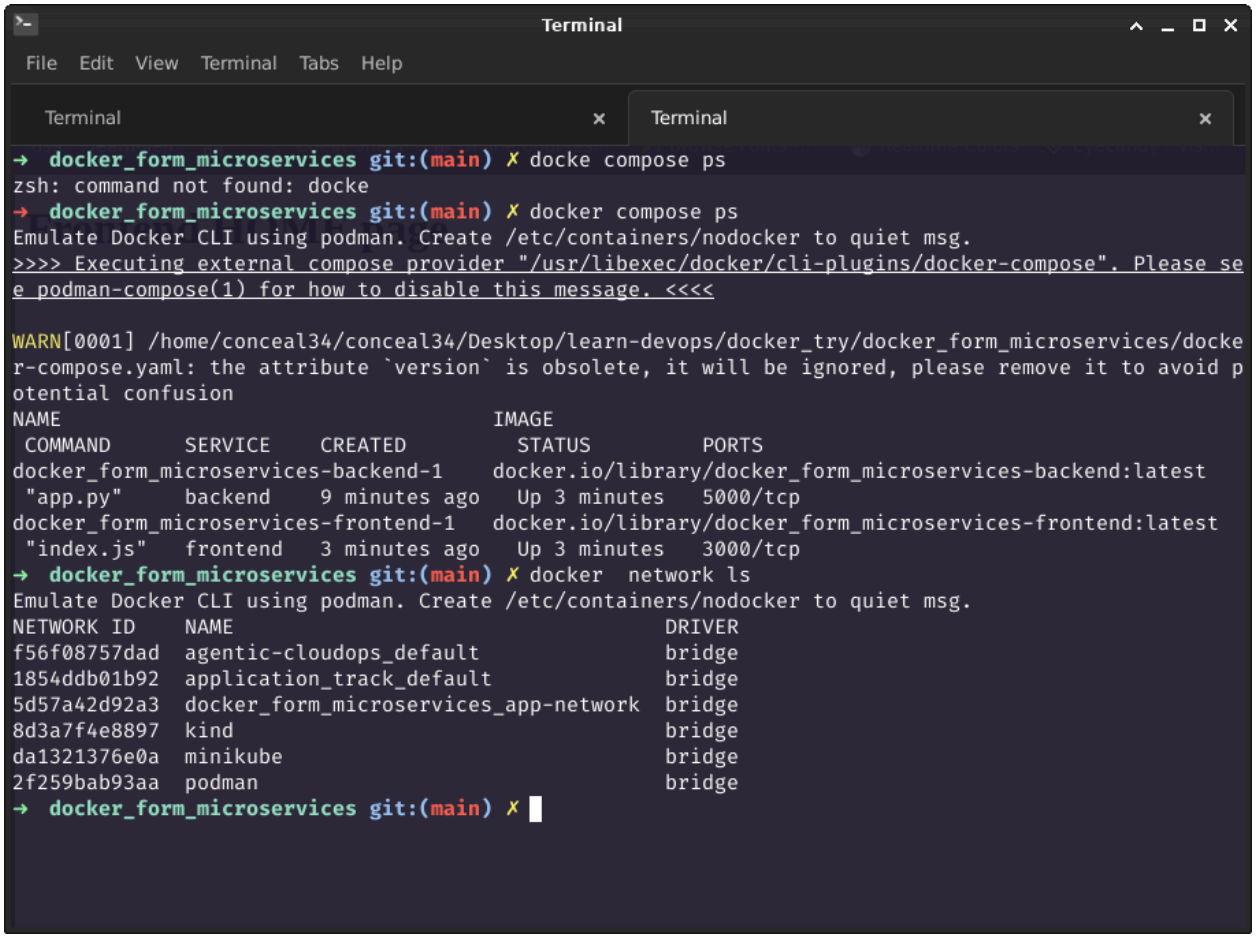
8. Upload both images to Docker Hub, and push your whole code to GitHub. Add node_modules and other non-required files (e.g., `.vscode`) to .gitignore.




Terminal
File Edit View Terminal Tabs Help
Terminal
Terminal
--> 2b2d03a524ce
STEP 6/7: CMD ["index.js"]
--> Using cache 27c628edcc8b2a1003e95aec09c4685446a3156ec2346d8be271703379a9de99
--> 27c628edcc8b
STEP 7/7: LABEL "com.docker.compose.image.builder"="classic"
--> Using cache 4b8a83602251a6c4f8581e457dc197304c91d9259246cbcc4f8fb1c327ccee3d
COMMIT docker.io/library/docker_form_microservices-frontend:latest
--> 4b8a83602251
Successfully tagged docker.io/library/docker_form_microservices-frontend:latest
4b8a83602251a6c4f8581e457dc197304c91d9259246cbcc4f8fb1c327ccee3d
Successfully built 4b8a83602251
Successfully tagged docker_form_microservices-frontend
[+] Running 0/1
Container docker_form_microservices-frontend-1 Created 2.2s
Attaching to backend-1, frontend-1
frontend-1 | Frontend service is running on port 3000
backend-1 | * Serving Flask app 'app'
backend-1 | * Debug mode: on
backend-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
backend-1 | * Running on all addresses (0.0.0.0)
backend-1 | * Running on http://127.0.0.1:5000
backend-1 | * Running on http://10.89.3.2:5000
backend-1 | Press CTRL+C to quit
backend-1 | * Restarting with stat
backend-1 | * Debugger is active!
backend-1 | * Debugger PIN: 276-187-126
backend-1 | 10.89.3.2 - - [04/Nov/2026 07:33:49] "GET / HTTP/1.1" 200 -
backend-1 | 10.89.3.2 - - [04/Nov/2026 07:33:51] "GET /favicon.ico HTTP/1.1" 404 -


docker_form_microser...
localhost:5000
Document
localhost:3000
Relaunch to update
dsa
learn-self
Cheat Sheet
Online Courses...
Browse Fonts -...
Realtime Colors
Eyecandy - Vis...
ProductionCrat...
Free Music For...
CinePacks
Crate - Expand...
Codrops | Keepi...
All Bookmarks
Frontend HOME page


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localhost:5000
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Realtime Colors
Eyecandy - Vis...
ProductionCrat...
Free Music For...
CinePacks
Crate - Expand...
Codrops | Keepi...
All Bookmarks
Tree
Chart
JSON Editor
{
"message": "Welcome to the BACKEND!"
}


Terminal
File Edit View Terminal Tabs Help
Terminal
Terminal
→ docker_form_microservices git:(main) ✗ docke compose ps
zsh: command not found: docke
→ docker_form_microservices git:(main) ✗ docker compose ps
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
>>>> Executing external compose provider "/usr/libexec/docker/cli-plugins/docker-compose". Please see podman-compose(1) for how to disable this message. <<<<
WARN[0001] /home/conceal34/conceal34/Desktop/learn-devops/docker_try/docker_form_microservices/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME IMAGE
COMMAND SERVICE CREATED STATUS PORTS
docker_form_microservices-backend-1 docker.io/library/docker_form_microservices-backend:latest
"app.py" backend 9 minutes ago Up 3 minutes 5000/tcp
docker_form_microservices-frontend-1 docker.io/library/docker_form_microservices-frontend:latest
"index.js" frontend 3 minutes ago Up 3 minutes 3000/tcp
→ docker_form_microservices git:(main) ✗ docker network ls
Emulate Docker CLI using podman. Create /etc/containers/nodocker to quiet msg.
NETWORK ID NAME DRIVER
f56f08757dad agentic-cloudops_default bridge
1854ddb01b92 application_track_default bridge
5d57a42d92a3 docker_form_microservices_app-network bridge
8d3a7f4e8897 kind bridge
da1321376e0a minikube bridge
2f259bab93aa podman bridge
→ docker_form_microservices git:(main) ✗